

CONSTRA

An integrated geoscience workspace for engineering and
construction

Vision & Initial Scope

Draft v0.2 · 2026-04-26

*Working name: **Constra** (subject to change)*

1. Executive Summary

Constra is a desktop application for **engineering geoscience** — the daily working environment for engineering geologists and engineering geophysicists in geotechnical site investigation, civil and construction engineering, mining engineering, and environmental / engineering hydrogeology. It is built around three linked views — Map, Section, and 3D — that share a single real-world coordinate frame, so the boreholes, in-situ tests, near-surface geophysics, topographic data, and CAD-derived plans that drive an engineering project all live together in one project.

The oil & gas industry has had this kind of integrated, coordinate-aware workspace for decades — Petrel, by Schlumberger, is the canonical example and a genuinely outstanding product within its domain. Engineering geoscience has had no equivalent. Practitioners assemble cross-sections by hand in CAD, plot data in generic tools, and stitch results together across half a dozen single-purpose applications, with all the workflow friction and silent inconsistency that implies. Constra exists to close that gap — to give the engineering practitioner the same kind of integrated workspace that the oil industry has long enjoyed, but built specifically around engineering data, engineering coordinate systems, and the engineering interpretation workflow.

Petrel and Surfer appear in this document only as *exemplars*. Petrel illustrates what an integrated, multi-view, coordinate-aware workspace looks like when one is purpose-built for an industry. Surfer illustrates the limits of generic plotting tools that lack a project model. Neither is a competitor to Constra: Constra serves a different industry — one that currently has nothing of either kind purpose-built for it.

2. Exemplar — Petrel

Petrel is a flagship interpretation platform built by Schlumberger for the oil & gas industry. It is shown here only as an exemplar — a legendary product within its industry, used to illustrate what an integrated, coordinate-aware geoscience workspace looks like when one is purpose-built for a domain. Petrel does not serve engineering, and Construa is not a Petrel alternative; the two products live in different industries. The figures that follow are reproduced so that the reader can see, concretely, what the engineering equivalent — Construa — should feel like in use.

Within its own industry, Petrel integrates geophysical, geological, hydrogeological, and reservoir data into a single interactive environment where multiple data types can be combined and visualised together. Data is displayed in both 2D and 3D, and every view stays in sync with the others.

2.1 Map View

All datasets are displayed in their true geographic location inside a shared Map View. Satellite imagery, seismic line locations, well locations, and cultural features (roads, buildings, infrastructure) can all be layered in the same window.

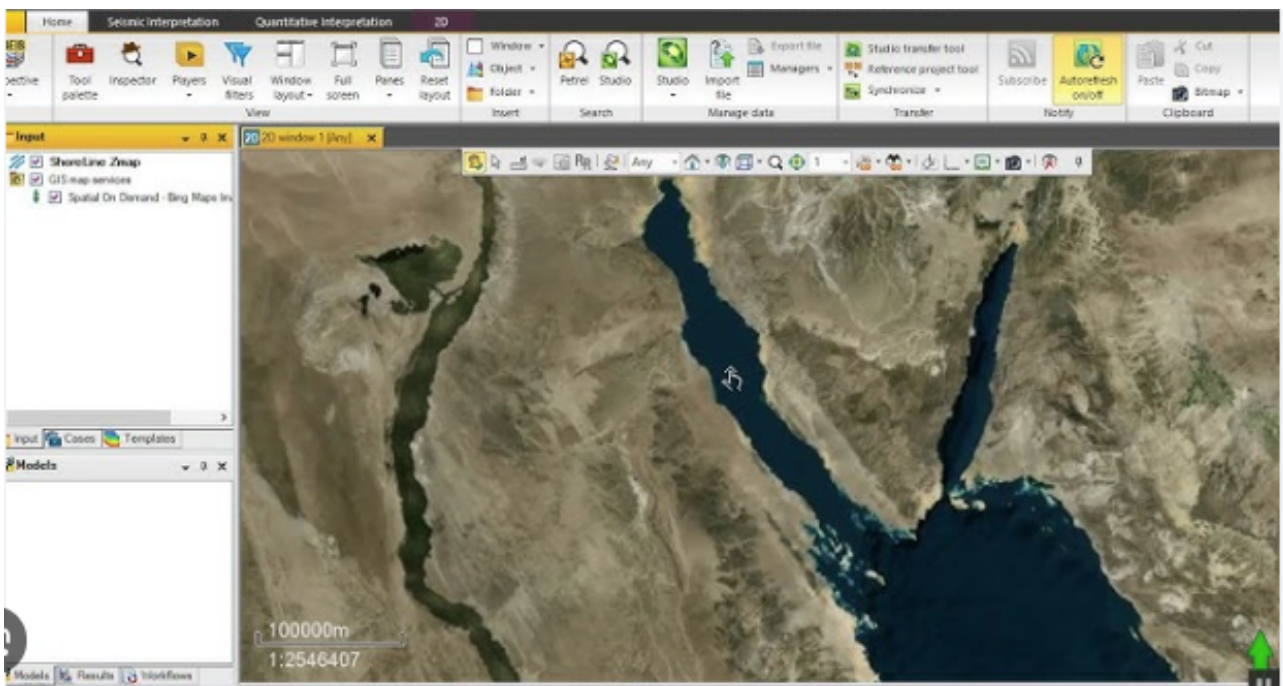


Figure 1 — Satellite basemap in Petrel's Map View. A satellite image of the project area is loaded as the bottom layer of the map, providing real-world geographic context for everything placed on top of it.

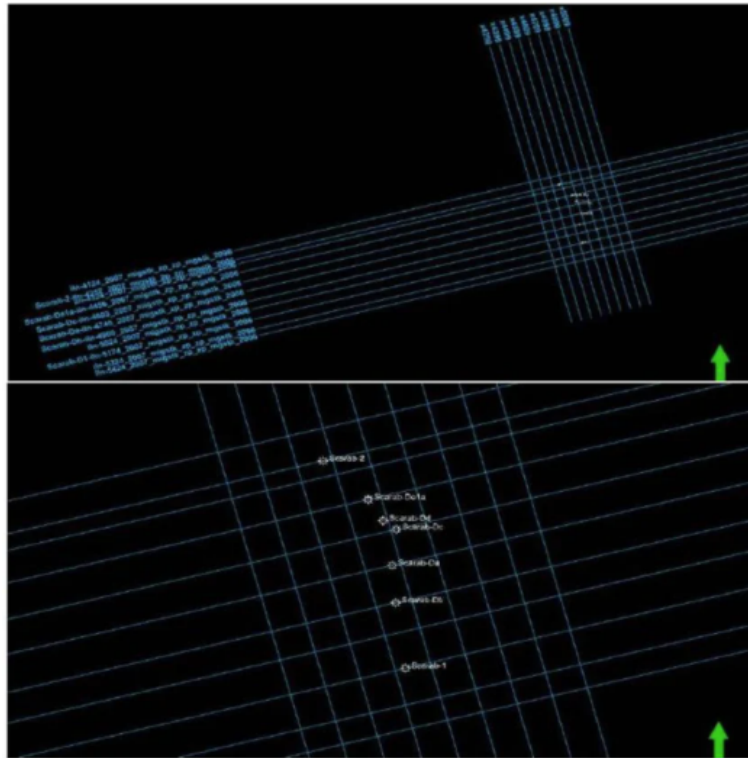


Figure 2 — Seismic lines overlaid with well locations. The Map View shows the geometry of acquired 2D seismic lines together with well locations, so an interpreter can pick which line to open next based on where the wells are.

2.2 Section View

The Section View is used to display data that is naturally viewed as a vertical slice through the earth — 2D seismic lines, well logs, horizons, and faults. Selecting a line on the map opens it here as a vertical section, ready for interpretation.

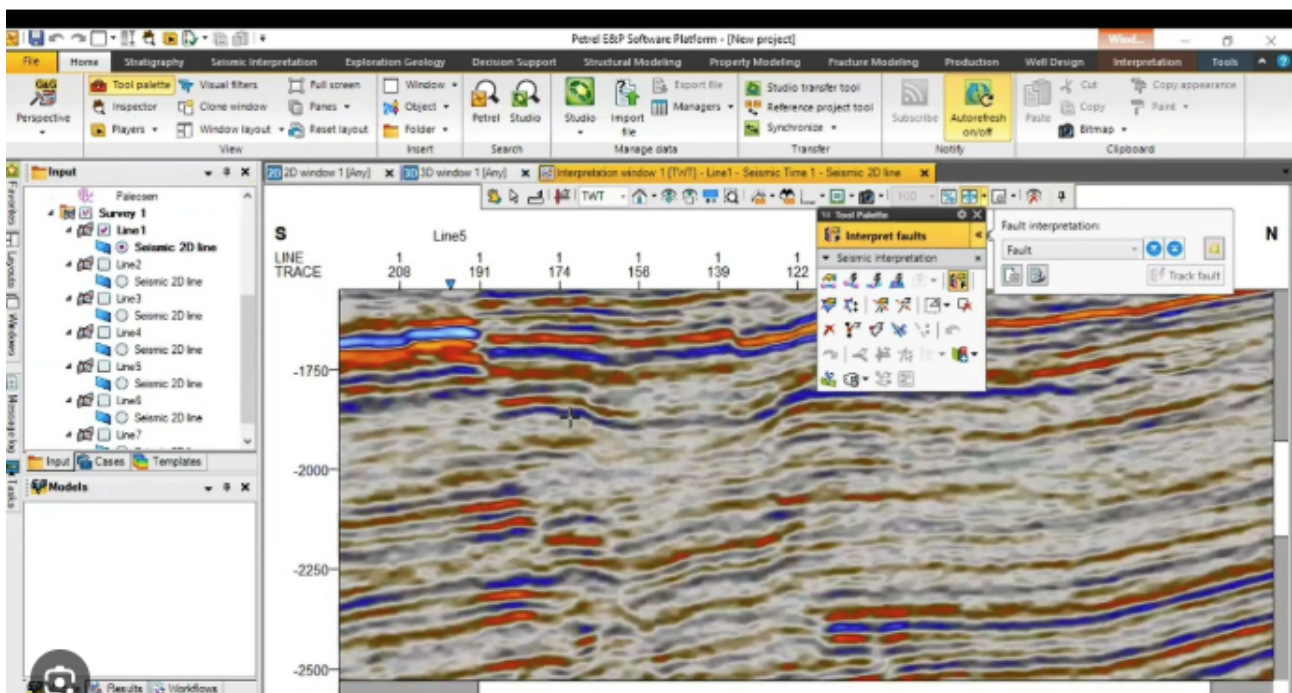


Figure 3 — 2D seismic line displayed as a vertical section in Petrel.

Key behaviour. Map View and Section View are fully interactive. An action taken in one view — selecting a line, picking a horizon, moving a crosshair — is immediately reflected in the other. The same is true for every other data type loaded into the project.

2.3 3D View

In addition to 2D, all data can be viewed, interpreted, and displayed in a 3D View, which is also synchronised with the Map View and Section View.

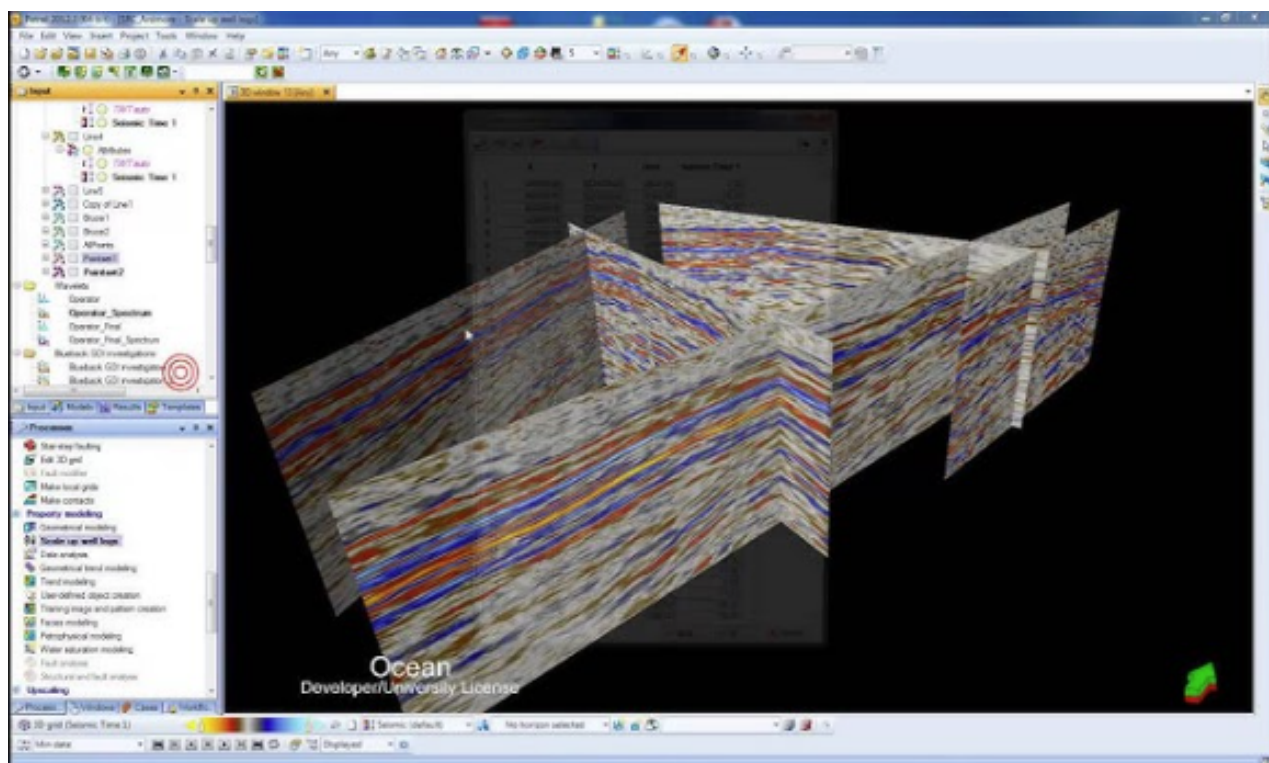


Figure 4 — 3D view of seismic lines in Petrel. Multiple 2D seismic lines are rendered together in three-dimensional space, making it easier to see how the subsurface looks along intersecting acquisition directions.

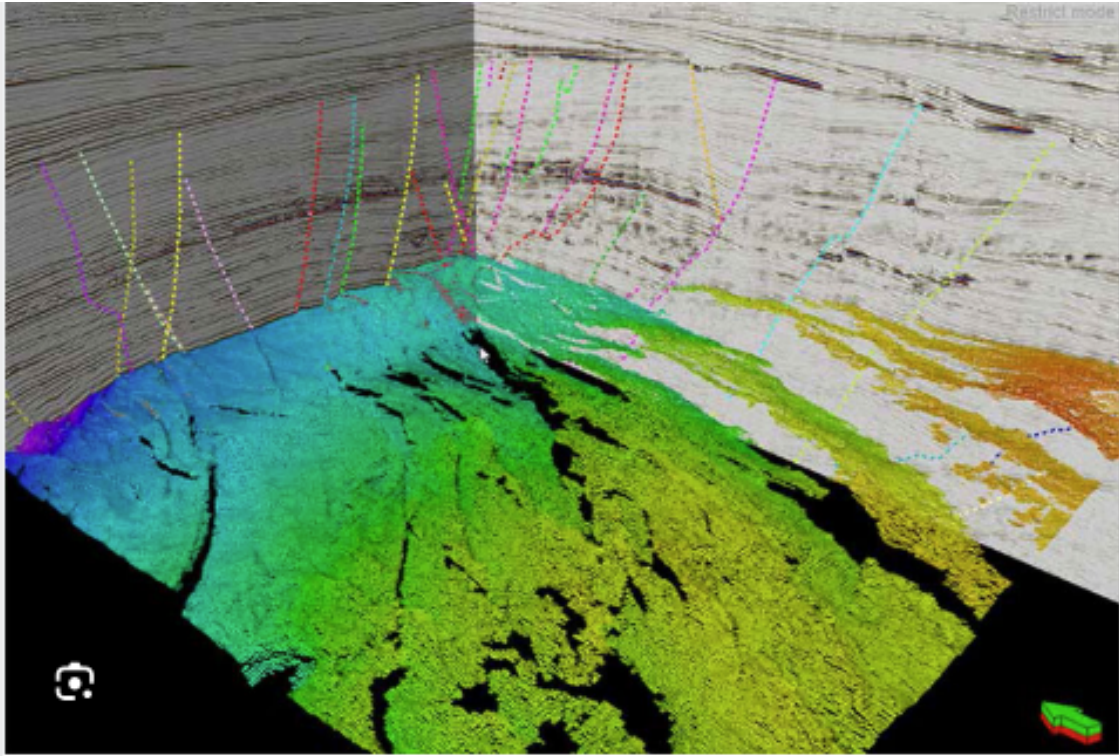


Figure 5 — 3D view of reservoir data with a well. A reservoir model is visualised in 3D alongside a well trajectory, so the relationship between the well path and the reservoir body can be assessed directly.

2.4 Coordinate System

Petrel works natively in real-world coordinates, both geographic (latitude/longitude) and projected (Easting/Northing). Every dataset is placed in its true location, and projections are handled by the software rather than by the user. This is what makes the cross-view interactivity meaningful: a well clicked in the Map View is the *same* well in the Section View and the 3D View, because the underlying coordinate system is unified. Constra targets the same property for engineering data.

3. Exemplar — Surfer

Surfer (by Golden Software) is a widely used plotting and contouring package for spatial data. It is shown here as an exemplar of a different category — a generic plotting tool, not an industry-specific workspace — to illustrate where the integration limit sits when a tool is not built around a project model. Surfer is good at what it does; the point is simply that what it does is not what an engineering practitioner needs as a daily working environment.

3.1 No dedicated views

Surfer does not provide separate Map, Section, and 3D views that share state. All output — whether conceptually a map or a vertical section — is drawn into the same generic plot window. There is no shared geographic canvas, no automatic basemap layer, and no cross-view synchronisation. Each figure is a standalone artefact constructed by hand.

3.2 No project-level coordinate frame

Surfer does not place datasets into a shared real-world projected coordinate frame. Data is plotted against whatever coordinate values are supplied in the input file, and the user is responsible for ensuring those values are consistent across datasets. There is no built-in projection engine that unifies every dataset into a single geographic reference.

3.3 Manual workflow

Every dataset, every object, every label is added to the plot by the user. This is acceptable when producing a single figure, but it scales poorly when many datasets must be kept consistent, when data is updated frequently, or when an interpreter needs to move fluidly between map, section, and 3D perspectives of the same subsurface model.

Product page: <https://www.goldensoftware.com/products/surfer>

4. The Engineering-Geoscience Gap

The point of contrasting these two exemplars is not to compare them with each other, but to show what the engineering practitioner has access to today. The oil & gas industry has Petrel; everyone, including engineering teams, has Surfer-class plotting tools; engineering geoscience has no widely available integrated workspace of its own. The table below makes that gap explicit.

Capability	Petrel (oil & gas)	Surfer (generic plotting)	Engineering geoscience today
Industry-specific integrated workspace	Yes	—	None widely available
Single project / shared coordinate frame	Yes	No	None widely available
Real-world coordinates, automatic reprojection	Yes	No	Handled per-tool, per-file
Linked Map / Section / 3D views	Yes	No	None widely available
Multiple simultaneous cross-sections	Yes	No	Manual in CAD
Native readers for the industry's primary data	Yes (oil & gas formats)	—	Fragmented across single-purpose tools
Cross-view selection / propagation	Yes	No	None widely available

Constra is built to fill the rightmost column.

5. Problem Statement & Target User

5.1 The problem

Engineering projects — foundations, slopes, tunnels, dams, roads, mining works, environmental sites — generate large quantities of geological and geophysical data: borehole logs, in-situ tests (CPT, SPT, and others), near-surface geophysics surveys (refraction, MASW, GPR, ERT), topographic and bathymetric data, CAD-derived plans, and the engineer's own interpreted cross-sections. Today this data is handled across a fragmented toolchain — a CAD package for plans and sections, a plotting tool for graphs, a GIS for spatial layers, and Excel for tabular records. There is no integrated workspace, comparable to what oil & gas has in Petrel, that brings these data into one project, in one coordinate system, with linked Map, Section, and 3D views.

Constra exists to be that workspace.

5.2 Who Constra is for

Constra is built specifically for the **engineering geologist** and the **engineering geophysicist** — the practitioner who produces interpreted geological and geophysical models of the near surface and has to communicate them to civil designers, contractors, regulators, and clients. The primary domains of practice are:

- Geotechnical site investigation.
- Civil and construction engineering — foundations, slopes, dams, tunnels, roads, bridges, embankments.
- Mining engineering and exploration that does not fit the oil & gas paradigm.
- Environmental engineering and engineering hydrogeology.

The shared characteristic across all four is that the practitioner is working in real-world coordinates, with multiple data types per project, producing interpreted outputs (cross-sections, maps, 3D models) that other engineers and stakeholders consume.

5.3 What Constra is not

- **Not a numerical solver.** Constra is an interpretation and visualisation environment — it does not run finite-element analyses, slope-stability calculations, groundwater flow models, or seismic processing pipelines. It produces and displays interpreted models that those tools can consume.
- **Not (yet) a web application.** v0.1 is a desktop application for Windows. The chosen technology stack does not preclude a future web or cross-platform release, but neither is committed at v0.1.
- **Not a multi-user collaboration platform.** A single practitioner working on a single project is the v0.1 case. Multi-user editing, cloud sync, and shared review are explicitly future work.

6. Initial Release Scope (v0.1)

Constra is meant to be a sophisticated, ambitious tool — the engineering industry's equivalent of a Petrel-class workspace, not a minimalist proof-of-concept. The v0.1 scope below is the first coherent release subset: enough to be genuinely useful on a real engineering project, while leaving room for everything else to follow.

6.1 In scope for v0.1

- **Project container.** A Constra project is a folder holding data sources plus a project-level coordinate reference system (CRS) and project metadata.
- **Map View.** 2D canvas that renders all datasets in their real-world projected coordinates, with pan, zoom, and layer visibility controls. The user can lay out **multiple cross-section corridors** (straight or polyline) on the map for use by the Section View.
- **Section View.** Vertical-section workspace supporting **multiple sections open simultaneously**, each linked to a corridor defined on the map. Each section displays the data that intersects its corridor — boreholes, in-situ test traces, near-surface geophysics data — in correct relative depth and along-corridor distance.
- **3D View.** A 3D scene rendering all loaded datasets in space, linked to the same selection model as Map View and Section View.
- **Engineering data import.** Native readers for the data formats primary to engineering geoscience workflows. The exact format set for v0.1 is being finalised; current candidates include borehole records (e.g. AGS), in-situ test data (CPT, SPT), near-surface geophysics (refraction, MASW, GPR, ERT), DEM/DTM rasters (GeoTIFF, ASCII grid), vector GIS layers (shapefile, GeoJSON), and CAD plans (DXF). **SEG-Y is not a v0.1 priority** — engineering geophysics is largely non-SEG-Y.
- **Coordinate system support.** Per-project CRS via PROJ (pyproj), with on-the-fly reprojection of every loaded dataset into the project frame.
- **Cross-view selection.** Selecting a borehole, station, or feature in any view highlights it in every other view.
- **Windows installer.** A one-click MSI or exe for Windows 10/11, produced as a GitHub release artefact.

6.2 Deferred past v0.1

The list below contains items that fit Constra's industry but are not scheduled for the first release. Oil & gas-specific features (reservoir modelling, horizon picking on 3D seismic volumes, SEG-Y attribute generation) are not on this list — they belong to a different industry and are simply out of scope.

- Edit / write-back of imported data files (v0.1 is read-only).
- Multi-user collaboration, cloud sync, and shared project review.
- Mac and Linux installers (the chosen stack supports them; v0.1 simply does not commit to producing or supporting them).
- Python scripting / plugin API for user extensions.

- Web or SaaS deployment.
- Automated report generation (PDF / DOCX export of interpreted sections and maps).

7. Technology Choice

7.1 Chosen stack

- **Language:** Python 3.10+
- **GUI framework:** PySide6 (Qt 6 bindings, LGPL)
- **3D and scientific visualisation:** VTK (the library behind ParaView and 3D Slicer), optionally via PyVista for ergonomics
- **Coordinate systems:** pyproj (bindings to PROJ)
- **Raster I/O:** rasterio (GeoTIFF, ASCII grid, etc.)
- **Vector I/O:** pyogrio or Fiona for shapefile / GeoJSON; geopandas for in-memory manipulation
- **CAD I/O:** ezdxf for DXF (a primary engineering interchange format)
- **Engineering-format parsers:** selection finalised during M1 planning — likely python-ags4 for AGS borehole data, plus dedicated or custom readers for CPT, refraction, MASW, GPR, ERT.
- **Packaging for Windows:** PyInstaller, producing a standalone installer
- **Testing:** pytest with pytest-qt for GUI tests
- **Linting / formatting:** ruff, black

7.2 Why this stack

This is the same toolchain that underpins ParaView and 3D Slicer, two of the most successful open-source scientific visualisation applications in existence. Every hard problem on Constr's roadmap — large raster display, 3D scene management, coordinate reprojection, vector and CAD interop — already has a mature, well-maintained Python library. Where engineering-specific parsers are needed, Python is also the easiest ecosystem to add them in. Iteration speed matters at this stage, when getting the interaction model right is more important than micro-optimised performance; performance-critical components can later be moved to C++ or Rust without changing the user-facing application.

7.3 Platform strategy

v0.1 targets **Windows 10/11** only, because that is where engineering practitioners work day-to-day. Because the chosen stack is cross-platform by construction, Linux and macOS builds are expected to come essentially for free once the Windows build is stable — but they are not a v0.1 commitment.

8. Initial Task List

The backlog below is organised into milestones. Each milestone is meant to end in a runnable application that does a little bit more than the previous one.

M0 — Project bootstrap (complete)

- Public GitHub repository at github.com/atifali-pm/constra, with MIT licence, README, and .gitignore.
- Python project skeleton (pyproject.toml, src/ layout, dev / docs extras).
- Ruff + black + pytest + pytest-qt configured, plus GitHub Actions CI on Python 3.10 / 3.11 / 3.12.
- Hello-world PySide6 main window that launches and closes cleanly.
- Docker development image with X11 forwarding for visual runs on Linux hosts.

M1 — Project container, CRS, and engineering-format selection

- Define the on-disk Constra project format (project folder + project.toml).
- Implement project create / open / save.
- Project-level CRS setting (pick EPSG code at project creation, stored in the project file).
- Wire pyproj for on-the-fly reprojection of any loaded dataset into the project CRS.
- Finalise the v0.1 set of engineering data formats and pick / write the parsers.

M2 — Map View

- Build the Map View widget (Qt Graphics View or a VTK 2D render window).
- Load a DEM / DTM raster (GeoTIFF, ASCII grid) and display it reprojected into the project CRS.
- Load a shapefile / GeoJSON vector layer and overlay it.
- Load a CSV of point data (e.g. borehole heads) and plot each point with a label.
- Load a DXF plan via ezdxf and render it as a layer.
- Pan / zoom / layer-visibility controls and a layer panel.
- Tool to draw **multiple cross-section corridors** on the map for use by the Section View.

M3 — Section View (multiple sections)

- Build the Section View widget supporting **multiple sections open concurrently**, one per defined corridor.
- For each section, project the data that intersects the corridor (boreholes, in-situ test traces, near-surface geophysics) into (distance, depth) coordinates and render it.
- Configurable colour map, gain, vertical exaggeration, and depth scale.
- Link Section View to Map View — selecting a section corridor, feature, or borehole in one view highlights it in the other.

M4 — 3D View

- Build the 3D View widget using VTK / PyVista.
- Render the basemap, boreholes, near-surface geophysics traces, and section corridors in 3D space.
- Wire cross-view selection so the 3D View shares a selection model with the Map View and Section View.

M5 — Native engineering data and the first installer

- Implement the engineering-format parsers chosen in M1 (e.g. AGS borehole data, CPT/SPT, near-surface geophysics formats).
- Display borehole logs alongside the section view.
- Package the application for Windows with PyInstaller.
- Publish a signed Windows installer as a GitHub release artefact.

Shipping M5 means Construa v0.1 is a real engineering-geoscience workspace: project-scoped, coordinate-aware, with linked Map / Section / 3D views and native readers for the data engineers actually use. It is not a tiny proof of concept — it is the first usable release of an ambitious tool, and the foundation everything else is built on.